4. (a) Data Modelling and Form Design

Aim

To design the data model and forms of the Smart Lost and Found Management System in Zoho Creator using suitable field types, mandatory fields and validations, and to review the submitted records in reports.

Theory

Data modelling means deciding which data is needed and how it is split into forms and fields. In Zoho Creator every form becomes a table and every field has a type (Name, Email, Phone, Address, Date, Single Line, Multi Line, Drop Down, Radio, URL, Lookup, Subform and others). Mandatory fields, fixed choices and Deluge validations keep the stored data clean.

For the lost and found system three forms were designed: Lost_Items for items that owners report lost, Found_Items for items that finders hand in, and Claim_Requests for ownership claims against found items.

Procedure

1. Open Zoho Creator and create the application Smart Lost and Found Management System.
2. Create the Lost_Items form with the fields Contact Name, Email, Item Name, Category (drop down), Description, Date Lost, Location Lost and Item Image.

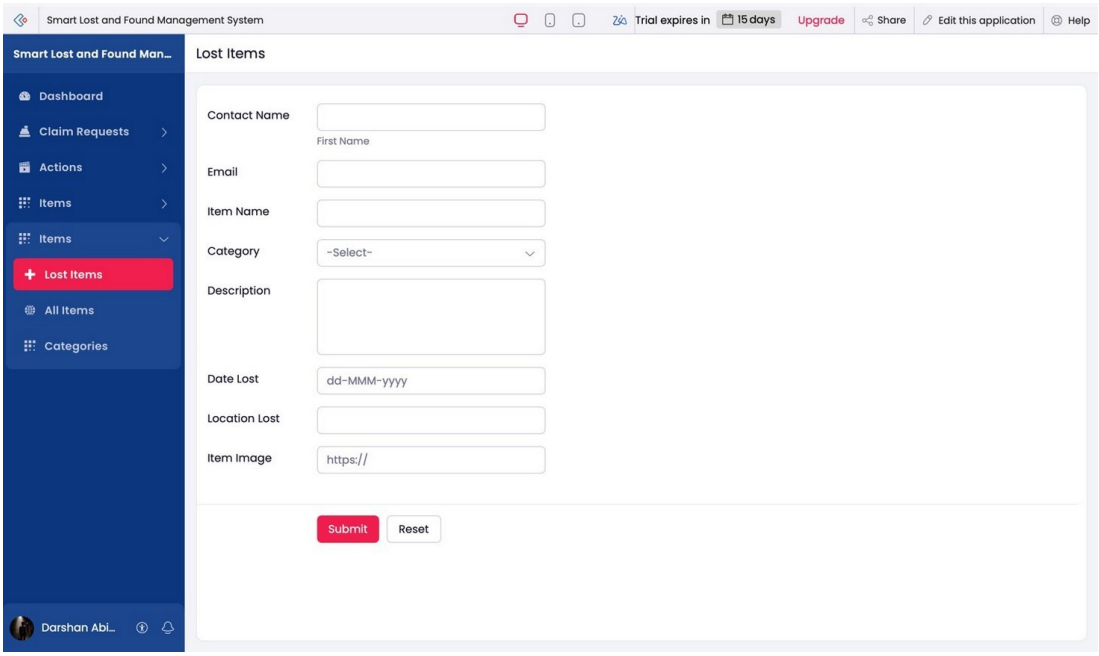


Fig. 4.1: Lost_Items form

3. Create the Found_Items form with Finder Name, Finder Email, Item Name, Category, Description, Date Found, Location Found, Item Image and a Status field with the choices Available, Claimed and Other.

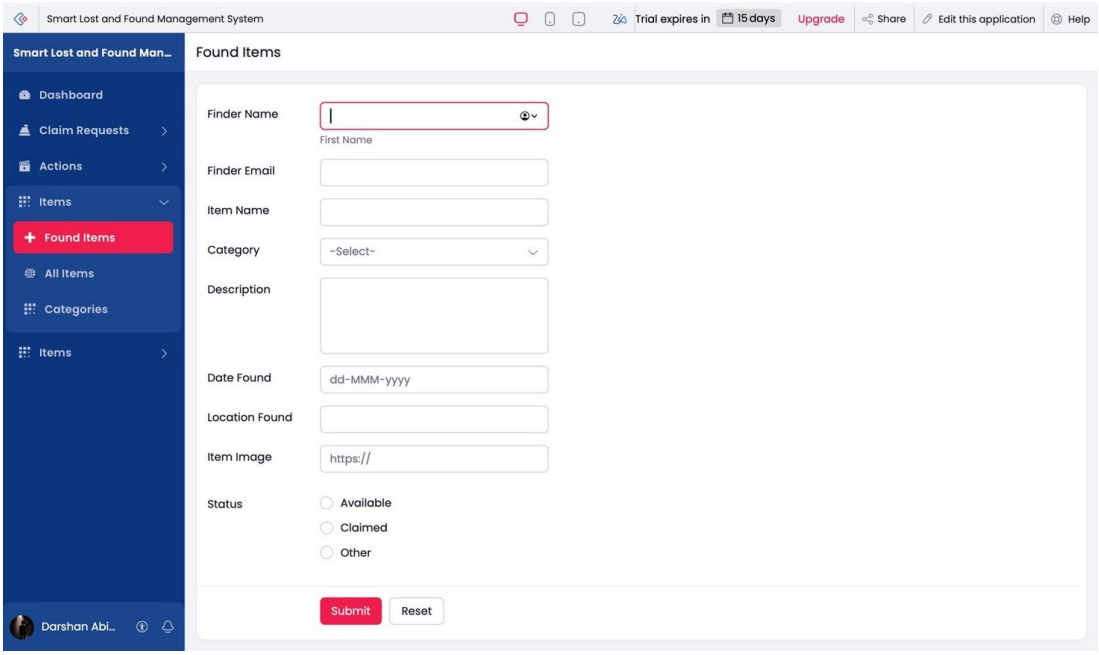


Fig. 4.2: Found_Items form

4. Create the Claim_Requests form with Claimant Name, Claimant Email, Found Item (lookup), Claim Date, Proof Ownership Details, Claim Status and Admin Remarks.

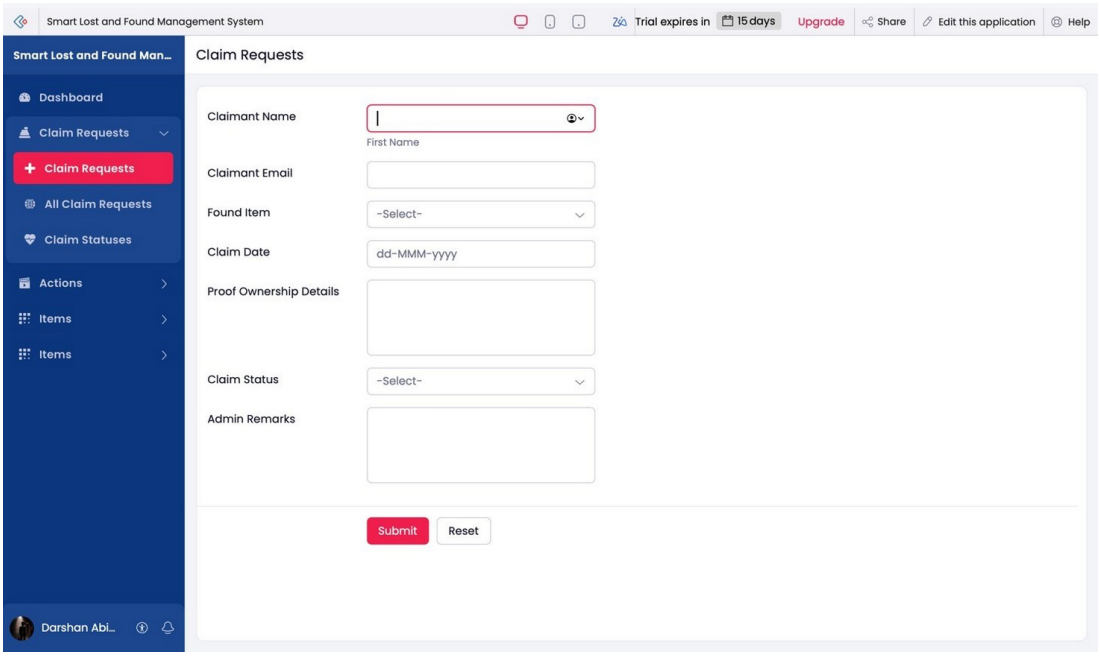
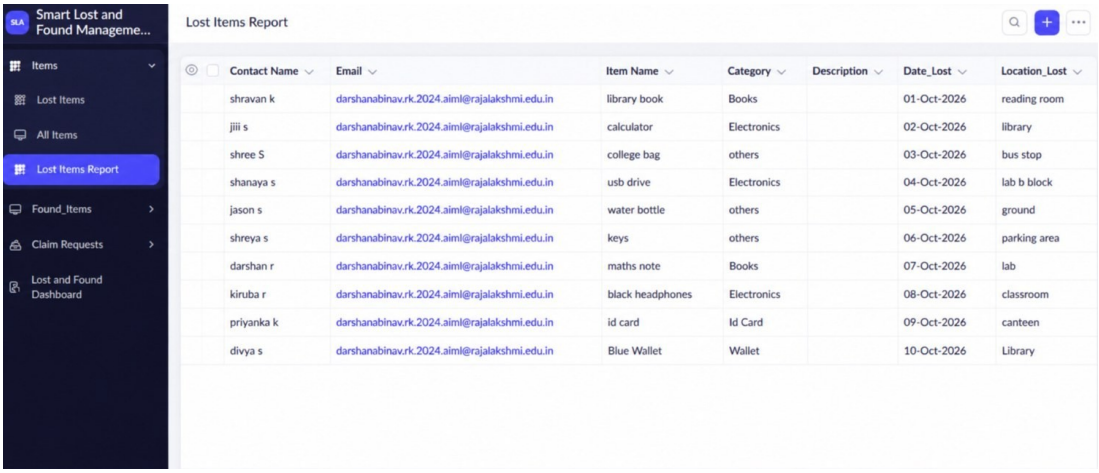


Fig. 4.3: Claim_Requests form

5. Mark the required fields, and add Deluge validations on the On Validate event so that a lost, found or claim date cannot be in the future and the same lost report cannot be entered twice (Experiment 6 gives the code).
6. Save the forms, open them in live mode, enter sample records and submit.
7. Open the reports to verify the submitted records.



Contact Name	Email	Item Name	Category	Description	Date_Lost	Location_Lost
shravan k	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	library book	Books		01-Oct-2026	reading room
jiii s	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	calculator	Electronics		02-Oct-2026	library
shree S	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	college bag	others		03-Oct-2026	bus stop
shanaya s	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	usb drive	Electronics		04-Oct-2026	lab b block
jason s	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	water bottle	others		05-Oct-2026	ground
shreya s	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	keys	others		06-Oct-2026	parking area
darshan r	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	maths note	Books		07-Oct-2026	lab
kiruba r	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	black headphones	Electronics		08-Oct-2026	classroom
priyanka k	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	id card	Id Card		09-Oct-2026	canteen
divya s	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	Blue Wallet	Wallet		10-Oct-2026	Library

Fig. 4.4: Lost Items Report showing the submitted lost items

Result

The forms of the Smart Lost and Found Management System were designed with suitable field types, mandatory fields and validations. Records submitted through the forms were stored and are displayed in the reports for review.